

Aluminium-Water Combustion for Zero Carbon Marine Shipping: Masters Proposal

Md Nadim Ahmed

December 12, 2025

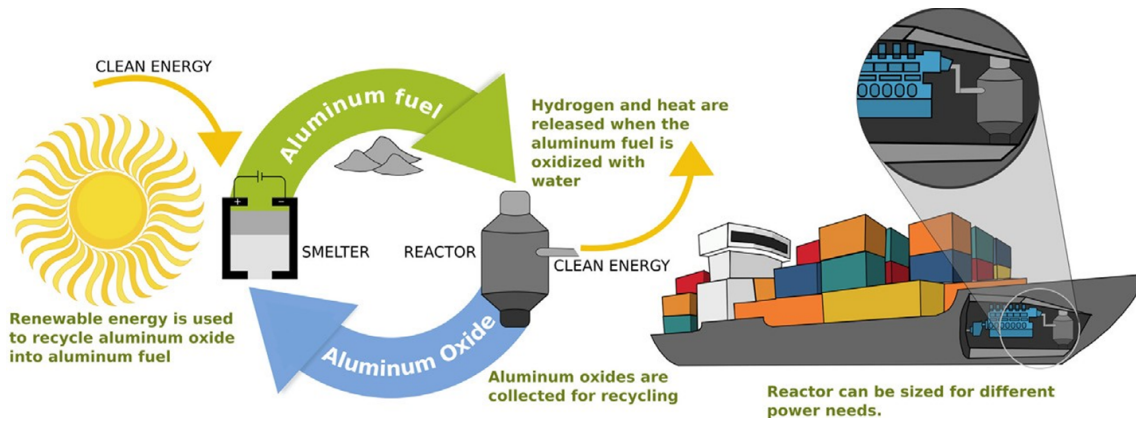


Figure 1: Aluminium powder as maritime shipping fuel (Energy Life-cycle). Adapted from [1]

Abstract

Table of Contents

List of Tables	4
List of Figures	4
1 Research Background	5
1.1 Preliminary Analysis of Electro-Fuels Options	5
2 Research Questions	7
3 Methodology	8
4 Proposed Project Timeline	9
5 References	10

List of Tables

List of Figures

1	Aluminium powder as maritime shipping fuel (Energy Life-cycle). Adapted from [1]	1
2	Falling cost of solar electricity vs Fossil Fuels [2]	5
3	Falling cost of lithium ion battery packs [3]	6

1 Research Background

Overview

Maritime shipping plays a crucial role in global trade, facilitating around 80% of all commerce [4]. This makes it essential for globalization, economic development, and maintaining living standards. However, the sector also contributes significantly to global greenhouse gas emissions, accounting for 2-3% of global emissions and approximately 10% of all transport-related emissions [5, 4].

In response, the International Maritime Organization (IMO) has set ambitious targets to curb emissions from international shipping. By 2030, the IMO aims to reduce annual greenhouse gas emissions by 20-30% compared to 2008 levels, with a more aggressive target of 70-80% reduction by 2040 [6].

1.1 Preliminary Analysis of Electro-Fuels Options

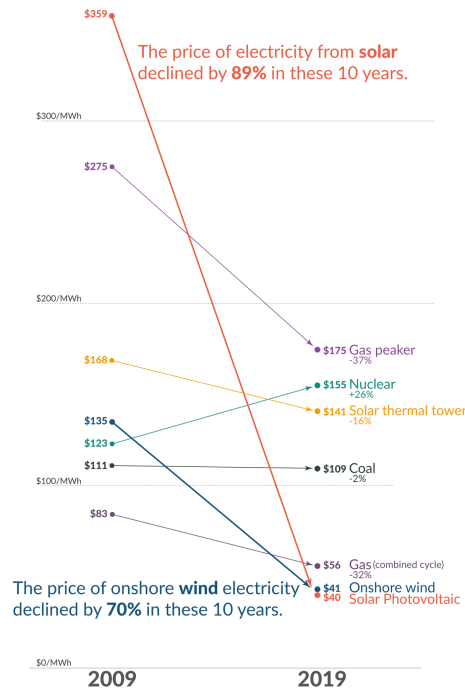


Figure 2: Falling cost of solar electricity vs Fossil Fuels [2]

Recent years have witnessed a dramatic decline in the cost of low-carbon electricity sources, particularly solar and wind power, as illustrated in Figure 2. This cost reduction has significantly enhanced the viability of transitioning to a low-carbon economy, yet substantial challenges persist in the so-called hard-to-abate sectors, among which marine shipping features prominently.

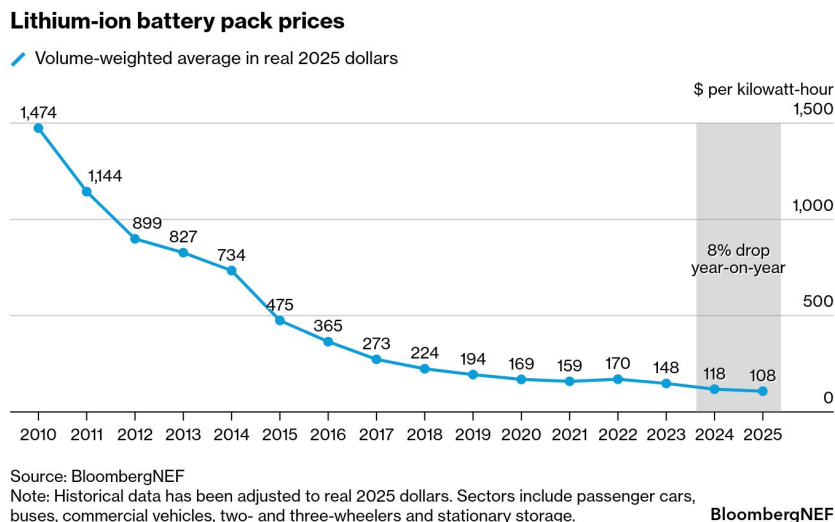


Figure 3: Falling cost of lithium ion battery packs [3]

The predominant method for storing electrical energy remains electrochemical batteries, with lithium-ion technology leading the charge. These batteries have increasingly displaced fossil fuels in passenger vehicles and, more recently, have begun penetrating short and medium-distance trucking applications [7]. This transformation has been driven largely by the precipitous decline in lithium battery prices, as demonstrated in Figure 3. The technology’s reach has even extended to inland shipping routes, where standardized and swappable battery packs are being deployed. Major battery manufacturers such as CATL are making substantial investments in these markets, signaling confidence in the technology’s continued expansion [9].

However, despite remarkable advances in both price and energy density, lithium-ion batteries remain largely impractical for deep-sea shipping, which accounts for approximately 70% of marine shipping emissions. The fundamental constraint lies in the current energy density limitations of lithium-ion technology, which ranges from 0.1 to 0.5 kWh/kg—substantially lower than gasoline, which can exceed 10 kWh/kg. This considerable disparity in energy density presents a formidable barrier to the electrification of long-distance maritime transport.

2 Research Questions

3 Methodology

4 Proposed Project Timeline

5 References

- [1] K. A. Trowell, S. Goroshin, D. L. Frost, and J. M. Bergthorson, “Aluminum and its role as a recyclable, sustainable carrier of renewable energy,” *Appl. Energy*, vol. 275, p. 115112, Oct. 2020.
- [2] M. Roser, “Why did renewables become so cheap so fast?,” *Our World in Data*, 2020. <https://ourworldindata.org/cheap-renewables-growth>.
- [3] M. Maisch, “BNEF: Lithium-ion battery pack prices fall to \$108/kWh, stationary storage becomes lowest price segment - Energy Storage — ess-news.com.” <https://www.ess-news.com/2025/12/09/bnef-lithium-ion-battery-pack-prices-fall-to-108-kwh-stationary-storage-becomes-lowest-price-segment/#pid=1>. [Accessed 12-12-2025].
- [4] H. Ritchie, “Cars, planes, trains: where do co2 emissions from transport come from?,” *Our World in Data*, 2020. <https://ourworldindata.org/co2-emissions-from-transport>.
- [5] H. Ritchie, “Sector by sector: where do global greenhouse gas emissions come from?,” *Our World in Data*, 2020. <https://ourworldindata.org/ghg-emissions-by-sector>.
- [6] “2023 IMO Strategy on Reduction of GHG Emissions from Ships — imo.org.” <https://www.imo.org/en/OurWork/Environment/Pages/2023-IMO-Strategy-on-Reduction-of-GHG-Emissions-from-Ships.aspx>. [Accessed 24-10-2024].
- [7] C. McKerracher, “China’s Clean-Truck Surprise Defies the EV Slowdown Narrative — BloombergNEF — about.bnef.com.” <https://about.bnef.com/insights/clean-transport/chinas-clean-truck-surprise-defies-the-ev-slowdown-narrative/>. [Accessed 12-12-2025].
- [8] Z. Weili, “The world’s first PACK-powered battery-swapping ship has completed its final prototype trials. — World Ports Organization — worldports.org.” <https://www.worldports.org/the-worlds-first-pack-powered-battery-swapping-ship-has-completed-its-final-prototype-trials/>. [Accessed 12-12-2025].

- [9] M. P. Release, “Maersk and CATL Forge Global Strategic Partnership in Supply Chain Electrification and International Logistics — maersk.com.” <https://www.maersk.com/news/articles/2025/10/10/maersk-and-catl-forge-global-strategic-partnership-in-supply-chain>. [Accessed 12-12-2025].